Numerical Methods for Computer Science

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1 Introduction

1.1 Why Numercial Methods in Computer Science

Numerical Analysis (*Numerische Methoden*) focuses on developing efficient algorithms and accurate approximations to solve mathematical problems that cannot be solved analytically.

1.1.1 Approximation Strategies: Promises and Limitations

A central task in numerical methods is function approximation. Depending on the properties of the target function and the domain of interest, different representation bases are chosen:

- **Taylor Polynomial Approximation:** Local polynomial approximations rely on Taylor series expansions. While effective locally, Taylor polynomials require high-order derivatives and strict smoothness assumptions ($f \in C^{m+1}$). Furthermore, convergence is strictly local around an expansion point x_0.
- **Global Polynomial Spaces & Projection:** To obtain global approximations without requiring derivatives, one can select a finite-dimensional subspace (e.g., polynomials up to degree m) spanned by a chosen basis, such as monomials $\{1, x, x^2, \ldots, x^m\}$ or orthogonal polynomials. The best approximation is then found by projecting the function onto this subspace under a chosen norm (e.g., L^2-norm).
- **Fourier Approximation:** For periodic functions, Fourier series provide a global approximation by expanding $f \in L^2(0,1)$ in terms of complex exponentials:

$$f(t) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{2\pi i k t}, \quad \text{where } \hat{f}(k) = \int_0^1 f(t) e^{-2\pi i k t} dt$$

  Fourier approximations achieve rapid (exponential) convergence for smooth periodic functions without requiring explicit derivative calculations.

1.1.2 Core Topics in Numerical Methods

Numerical analysis encompasses several fundamental domains aimed at constructing scalable and stable algorithms:

- **Interpolation:** Constructing an approximating function that passes exactly through a given set of discrete data points (x_j, y_j).
- **Numerical Quadrature:** Efficiently evaluating definite integrals $\int_a^b f(t)dt$ using weighted sums at discrete node points.
- **Numerical Linear Algebra:** Solving large linear systems $A\underline{x} = \underline{b}$ efficiently and stably.
- **Linear Least Squares (*Lineare Ausgleichsrechnung*):** Finding optimal parameters by minimizing $\|A\underline{x} - \underline{b}\|_2$ for overdetermined systems.
- **Nonlinear Systems of Equations & Least Squares:** Finding roots $\underline{f}(\underline{x}^*) = \underline{0}$ or solving non-linear optimization problems $\min_{\underline{p}} \|f(\underline{p}, \cdot) - y\|_2$.

1.1.3 Key Design Goals in Numerical Methods

When designing numerical algorithms, the main objectives are:

1. **Efficiency & Time Complexity:** Minimizing computation time and floating-point operations.
2. **Memory Footprint:** Optimizing memory usage for large-scale calculations.
3. **Quantifiable Accuracy & Convergence:** Guaranteeing and bounding errors for convergent sequences.
4. **Generality & Stability:** Ensuring robustness against rounding errors and subtractive cancellation.

1.2 Rounding Errors

Definition 1.1. Absolute Error: $\|\tilde{x} - x\|$ with $\|\cdot\|$ a norm in $\mathbb{R}^n$.

Definition 1.2. Relative Error: $\frac{\|\tilde{x}-x\|}{\|\tilde{x}\|}$ for $\tilde{x} \neq 0$, with $\|\cdot\|$ a norm in $\mathbb{R}^n$.

1.2.1 Machine Numbers

Remark 1.3. In modern computer systems, numbers are represented as floating-point numbers using a mantissa-exponent representation.

Definition 1.4. The **Mantissa-Exponent** representation of a number is $\pm M \cdot B^E$, where M is the mantissa containing the significant digits and precision, $B \geq 2$ is the base (typically 2 or 10), and $E \in \{e_{\min}, \ldots, e_{\max}\}$ is the exponent determining the scale or magnitude of the number by shifting the radix point.

Example 1.5. Cancellation in Quadratic Equations: The polynomial $p(x) = (x - c)(x - 1/c) = x^2 - (c + 1/c)x + 1$ has roots c and $1/c$. When using the standard quadratic formula (*Mitternachtsformel*) to calculate these roots: The relative error for the smaller root $1/c$ is significantly larger than for the larger root c. This occurs because calculating the smaller root requires subtracting two nearly equal values ($-a$ and $\sqrt{D}$), leading to a loss of significant digits.

Code Example 1.6

```
D = a**2 - 4*b
wD = sqrt(D)
z = ((-a + wD)/2, (-a - wD)/2)
```

Definition 1.7. This phenomenon is called **cancellation** (*Auslöschung*). It describes how the subtraction of two nearly equal numbers dramatically amplifies relative errors, even though the error of the subtraction operation itself is negligible.

Remark 1.8. Stable Alternative via Vieta's Formulas: To avoid cancellation when solving $x^2 + ax + b = 0$, compute the numerically stable larger root x_2 first, and obtain the smaller root x_1 using $x_1 x_2 = b \implies x_1 = b/x_2$:

Code Example 1.9

```
x2 = (sqrt(D) - a)/2 if a < 0 else (-sqrt(D) - a)/2
x1 = b / x2
```

1.2.2 Numerical Differentiation

Remark 1.10. We can use the forward difference quotient to approximate local derivatives:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Code Example 1.11

```
f = exp; df = exp; x = 0; h = 0.1
for k in range(1, 21):
    ndf = (f(x+h) - f(x)) / h
    h *= 0.1
```

$\log_{10}(h)$	relativer Fehler
-1	0.05170918075648
-2	0.005016708416 79
-3	0.000500166708 38
-4	0.000050001667 14
-5	0.000005000006 96
-6	0.000000499962 18
-7	0.000000049433 68
-8	0.000000006077 47
-9	0.000000008274 037
-10	0.000000008274 037
-11	0.000000008274 037
-12	0.000088900582 34
-13	0.000799277837 36
-14	0.000799277837 36
-15	0.110223024625 16
-16	1.000000000000 00

Remark 1.12. Observe that the approximation error initially decreases as h grows smaller, reaching optimal accuracy around $h \approx \sqrt{2 \cdot \text{eps}} \approx 10^{-8}$. However, as $h \to 0$ further, cancellation errors dominate because $f(x+h) \approx f(x)$, causing total accuracy to rapidly degrade.

Theorem 1.13. Complex Step Derivative: If f is analytic, we can use a complex step to perform numerically stable differentiation without cancellation:

$$f(x_0 + ih) = f(x_0) + ihf'(x_0) - \frac{h^2}{2}f''(x_0) - \frac{ih^3}{6}f'''(x_0) + \mathcal{O}(h^4)$$

Taking the imaginary part yields:

$$\text{Im}\left(f(x_0 + ih)\right) = hf'(x_0) - \frac{h^3}{6}f'''(x_0) + \mathcal{O}(h^5)$$

$$\implies f'(x_0) = \frac{\text{Im}\left(f(x_0 + ih)\right)}{h} + \mathcal{O}(h^2) \approx \frac{\text{Im}\left(f(x_0 + ih)\right)}{h}$$

Since this formulation involves no subtraction of function values, it is immune to subtractive cancellation for arbitrarily small values of h.

1.2.3 Convergence Acceleration via Richardson Extrapolation

Theorem 1.14. Symmetric Difference Quotient and Error Expansion: Using the central difference quotient $d(h) = \frac{f(x+h)-f(x-h)}{2h}$, Taylor expansion yields an error expansion containing only even powers of h:

$$f'(x) - d(h) = c_1h^2 + c_2h^4 + c_3h^6 + \cdots + \mathcal{O}(h^{2n+2})$$

Theorem 1.15. Richardson Extrapolation: By combining approximations evaluated at step sizes h and $h/2$, the leading error term c_1h^2 can be eliminated:

$$d_1(h) = \frac{4d(h/2) - d(h)}{3}$$

The refined approximation $d_1(h)$ achieves an increased convergence order of $\mathcal{O}(h^4)$:

$$f'(x) - d_1(h) = \tilde{c}_2h^4 + \tilde{c}_3h^6 + \dots$$

Definition 1.16. Richardson Scheme: Higher-order accuracy can be computed recursively using the triangular scheme ($R_{l,k}$ table) for levels $l = 0, 1, 2, \dots$ and extrapolation steps $k = 1, 2, \dots$:

$$R_{l,0} = d\left(\frac{h}{2^l}\right), \quad R_{l,k} = \frac{4^kR_{l,k-1} - R_{l-1,k-1}}{4^k - 1}$$

Each column k increases the order of convergence such that:

$$f'(x) - R_{l,k} = \mathcal{O}\left(\left(\frac{h}{2^l}\right)^{2k+2}\right)$$

This allows reaching high numerical precision at significantly larger step sizes h, effectively suppressing rounding and cancellation errors.

1.3 Matrices & Compute-Work

Definition 1.17. In NumCS we define the Compute-Work as the theoretical, asymototic number of elementary operations.

Remark 1.18. Observe that this definition has (almost) noting to do with the actual compute-time on a chip.

Example 1.19. Worst case Compute-Work for Gauss-Elimination

- General: $O(n^3)$
- Upper Triangular: $O(n^2)$
- Diagonal: $O(n)$

Remark 1.20. Numpy does not automatically adjust algorithms to the given inputs. We have to manually exploit the structure of ex. matrices and vectors.

Example 1.21. Let D be a diagonal matrix and let A be some other matrix. Using $np.dot(D, A)$ will result in $O(n^3)$ runtime just like any normal matrix multiplication. We have to manually exploit the structure of D by broadcasting the diagonal entries to a vector and then use numpy to multiply this vector by each row.

Example 1.22. Let $R = triu(AB)$ be the uppertringular matrix of the matrix product of $A, B \in R^{p,n}$ calculating $y = Rx$ takes $O(pn^2)$ using a standard numpy implementation

$$y = np.triu(A@B.T)@x$$

Let $p = 1$ we can represent rewrite this to $A'@T@B'@x$ where A', B' are diagonal matrices made from the rank one matrices A, B, T is a uppertringular matrix filled with ones. Observe that these products can are just (reverse) partialsums we can calculate using cumulative sum in numpy in $O(n)$. We can use this concept for any p and get a runtime of $O(pn)$.

2 Interpolation

Definition 2.1. Interpolation: Given exact values

$$y_1 = f(x_1, x_2 \dots), y_2 = f(x_i, x_{i+1} \dots) \dots$$

of an unknonw function f determine an interplant $\tilde{f}$ such that

$$\tilde{f}(x_1, x_2, \dots) = y_1$$

.

Remark 2.2. We assume f to have specific properties (smooth, monotone, etc.) $f \in V$ and choose a Subspace $\tilde{f} \in V_n$ together wiht a basis $b \in V_n$

Example 2.3. If we choose V_n as the set of all polynomials of degree $\leq n$ we are calculating a polynomial interpolation of f.

Remark 2.4. Different spaces and basis are possible and used for different applications.

Example 2.5. Using $n+1$ datapoints we can solve the linear system of equations of the $n+1$ datapoints to get a polynomial of degree n.

$$a_n x_1^n + a_{n-1} x_1^{n-1} \dots a_0 = y_1$$

$$a_n x_2^n + a_{n-1} x_2^{n-1} \dots a_0 = y_2$$

$$\dots$$

$$a_n x_{n+1}^n + a_{n-1} x_{n+1}^{n-1} \dots a_0 = y_{n+1}$$

Definition 2.6. The matrix A formed by this system of linear equations is called a **Vandermonde matrix**.

Remark 2.7. Solving this takes $O(n^3)$ and might come with unreasonably large rounding error.

2.1 Newton

Definition 2.8. Newton-Basis: The newton basis $N_n(x)$ is

$$\prod_{i=0}^{n-1}(x - x_i)$$

Theorem 2.9. Using the newton basis we can continously increase the degree of interpolation without having to solve the whole system of equations at once.

$$p(x_0) = y_0 \implies p_0(x) = y_0$$

$$p(x_0, x_1) = p_0, y_1 \implies p_1(x) = p_0(x) + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0)$$

$$p(x_0, x_1, x_2) = p_1, p_1, y_2 \implies p_2(x) = p_1(x) + \frac{\frac{y_2 - y_1}{x_2 - x_1} - \frac{y_1 - y_0}{x_1 - x_0}}{x_2 - x_0}(x - x_0)(x - x_1)$$

$$\dots$$

Remark 2.10. This can be interpreted as

$$p_i(x) = b_1 + b_2 N_1 + b_3 N_2 + \dots$$

where b_i is the i-th devided difference definied by the follwing recurrence relation

$$b_i = f[x_0, \dots x_i] = \frac{f[x_1, \dots x_i] - f[x_0, \dots x_{i-1}]}{x_i - x_0}$$

and $f[x_i] = y_i$

2.2 Lagrange

Definition 2.11. Lagrange-polynomials are polynomials of the form

$$l_i(x) = \prod_{j=0, j\neq i}^{n} \frac{x - x_j}{x_i - x_j}$$

Theorem 2.12. Observe that for such a largrange polynomial $l_i(x)$ we have

$$l_i(x) = 0 \forall x_j \text{with} j \neq i$$

and

$$l_i(x_i) = 1$$

Theorem 2.13. Because the because the above holds, given a set of datapoints $(x_1, y_1) \dots (x_n, y_n)$ we can construct a polynomial that fullfilles all of the n conditions

$$p_n(x) = \sum_{i=0}^{n} y_i l_i(x)$$

2.3 Barycentric Interpolation-formula

Definition 2.14. Barycentric weights: We define the barycentric weights w_i for a set of nodes $x_0, \dots, x_n$ as

$$w_i = \prod_{j=0, j\neq i}^{n} \frac{1}{x_i - x_j}$$

and the node polynomial $\ell(x)$ as

$$\ell(x) = \prod_{i=0}^{n} (x - x_i)$$

Theorem 2.15. Using these definitions, we can rewrite the Lagrange basis polynomials as $l_i(x) = \ell(x)\frac{w_i}{x-x_i}$. Substituting this into the standard Lagrange polynomial yields the **first form of the barycentric interpolation formula**:

$$p_n(x) = \ell(x) \sum_{i=0}^{n} \frac{w_i}{x - x_i} y_i$$

Theorem 2.16. Derivation of the second form: Because an interpolating polynomial reproduces constant functions exactly, interpolating the function $f(x) = 1$ (where $y_i = 1 \forall i$) gives:

$$1 = \ell(x) \sum_{i=0}^{n} \frac{w_i}{x - x_i}$$

Dividing the first form of the barycentric formula by this identity eliminates the node polynomial $\ell(x)$ and yields the **second (or true) barycentric interpolation formula**:

$$p_n(x) = \frac{\sum_{i=0}^{n} \frac{w_i}{x-x_i} y_i}{\sum_{i=0}^{n} \frac{w_i}{x-x_i}}$$

Remark 2.17. The second form is numerically highly stable. Once the weights w_i are precomputed (which takes $O(n^2)$), evaluating $p_n(x)$ at any new point x takes only

$O(n)$ operations. Furthermore, adding a new data point only requires updating the weights, which is much cheaper than recalculating the entire Lagrange polynomial.

2.4 Chebyshev

Remark 2.18. Using equidistant nodes for polynomial interpolation of high degree often leads to severe oscillations at the edges of the interval. This is known as **Runge's phenomenon**. To minimize this error, we must choose our interpolation nodes strategically.

Theorem 2.19. The error of polynomial interpolation is given by:

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^{n} (x - x_i)$$

for some ξ in the interpolation interval. Since we cannot control the $(n+1)$-th derivative of the unknown function f, we must minimize the maximum absolute value of the node polynomial $\prod_{i=0}^{n}(x - x_i)$.

Definition 2.20. Chebyshev nodes: The roots of the Chebyshev polynomials of the first kind are called Chebyshev nodes. On the standard interval $[-1, 1]$, the $n+1$ nodes are defined as:

$$x_i = \cos\left(\frac{2i+1}{2n+2}\pi\right) \quad \text{for } i = 0, 1, \ldots, n$$

Remark 2.21. If the interpolation interval is $[a, b]$ instead of $[-1, 1]$, the nodes can be shifted and scaled using an affine transformation:

$$x_i^{[a,b]} = \frac{a+b}{2} + \frac{b-a}{2} x_i$$

Theorem 2.22. Using Chebyshev nodes minimizes the maximum norm of the node polynomial on $[-1, 1]$:

$$\max_{x \in [-1,1]} \left| \prod_{i=0}^{n} (x - x_i) \right| = \frac{1}{2^n}$$

This node placement guarantees uniform convergence for sufficiently smooth functions and effectively suppresses Runge's phenomenon by clustering the nodes closer together at the edges of the interval.

2.5 Trigonometric Interpolation

Definition 2.23. Hilbert Space $L^2(0,1)$: The space of square-integrable complex-valued functions on the interval $[0,1]$ is defined as:

$$L^2(0,1) := \left\{ f : (0,1) \to \mathbb{C} \;\middle|\; \|f\|_{L^2(0,1)} = \sqrt{\langle f, f \rangle_{L^2(0,1)}} < \infty \right\}$$

equipped with the inner product $\langle g, f \rangle_{L^2(0,1)} := \int_0^1 \overline{g(t)} f(t)\, dt$.

Remark 2.24. L^2-Convergence: A function f is the L^2-limit of a sequence of functions $f_n \in L^2(0,1)$ if:

$$\lim_{n\to\infty} \|f - f_n\|_{L^2(0,1)} = 0 \iff \lim_{n\to\infty} \int_0^1 |f(t) - f_n(t)|^2\, dt = 0$$

Unlike pointwise convergence, L^2-convergence implies that the total energy (integrated squared difference) between f and f_n vanishes, allowing isolated point discrepancies.

Theorem 2.25. The complex exponential functions $\phi_k(x) = e^{2\pi i k x}$ for $k \in \mathbb{Z}$ form a complete orthonormal basis in $L^2(0,1)$:

$$\langle \phi_k, \phi_j \rangle_{L^2(0,1)} = \int_0^1 e^{-2\pi i k x} e^{2\pi i j x}\, dx = \begin{cases} 1, & \text{if } k = j \\ 0, & \text{if } k \neq j \end{cases}$$

Definition 2.26. Fourier Series: Any function $f \in L^2(0,1)$ can be represented by its Fourier coefficients $\hat{f}(k)$ as:

$$f(t) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{2\pi i k t}$$

where the Fourier coefficients $\hat{f}(k)$ are given by the orthogonal projection onto the basis functions:

$$\hat{f}(k) = \int_0^1 f(t) e^{-2\pi i k t}\, dt$$

Remark 2.27. Intuition & Physical Meaning: Analogous to decomposing a 3D vector into orthogonal coordinate axes, $f(t)$ is decomposed into pure periodic waves $e^{2\pi i k t}$ of integer frequency k. The magnitude $|\hat{f}(k)|$ represents the amplitude of frequency k, while its angle represents phase shift. Parseval's identity guarantees

energy conservation:

$$\sum_{k=-\infty}^{\infty} |\hat{f}(k)|^2 = \|f\|^2_{L^2(0,1)}$$

Remark 2.28. Smoothness and Decay Rate: The decay rate of Fourier coefficients depends directly on the smoothness of f. If $f^{(n)}$ is integrable ($\int_0^1 |f^{(n)}(t)|\,dt < \infty$), then:

$$\hat{f}(k) = \mathcal{O}(k^{-n})$$

Smoother functions yield faster coefficient decay, leading to rapid convergence of the Fourier series.

Remark 2.29. Derivative of Fourier Coefficients: The Fourier coefficients of the n-th derivative $f^{(n)}$ satisfy:

$$\widehat{f^{(n)}}(k) = (2\pi i k)^n \hat{f}(k)$$

This converts linear differential operators in the time domain into simple algebraic multiplications in the frequency domain.

Remark 2.30. Gibbs Phenomenon: At jump discontinuities, the truncated Fourier series overshoots near the edge by approximately 9% of the jump size. Increasing the number of terms shifts the overshoot closer to the discontinuity but does not eliminate its amplitude.

Definition 2.31. N-th Root of Unity: The primitive N-th root of unity is defined as:

$$\omega_N := \exp\left(\frac{-2\pi i}{N}\right) = \cos\left(\frac{2\pi}{N}\right) - i\sin\left(\frac{2\pi}{N}\right)$$

The roots of unity satisfy the discrete orthogonality relation:

$$\sum_{k=0}^{N-1} \omega_N^{kj} = \begin{cases} N, & \text{if } j \equiv 0 \pmod N \\ 0, & \text{if } j \not\equiv 0 \pmod N \end{cases}$$

This discrete orthogonality serves as the exact algebraic analogue of the continuous inner product of complex exponentials.

Definition 2.32. Discrete Fourier Transform (DFT): Given N equidistant nodes $x_l = \frac{l}{N}$ for $l = 0, \ldots, N-1$, approximating the continuous Fourier coefficients

$\hat{f}(k)$ via the trapezoidal rule yields the DFT:

$$\hat{f}(k) \approx \hat{f}_N(k) = \frac{1}{N} \sum_{l=0}^{N-1} f\left(\frac{l}{N}\right) \omega_N^{kl}$$

The transformation mapping samples $y = (f(x_0), \ldots, f(x_{N-1}))^T \mapsto c = F_N y$ is performed via the Fourier matrix $F_N = (\omega_N^{ij})_{i,j=0}^{N-1}$.

Remark 2.33. Fast Fourier Transform (FFT): Using a divide-and-conquer strategy (e.g., Cooley-Tukey algorithm), the matrix-vector multiplication $F_N y$ can be evaluated in $\mathcal{O}(N \log N)$ operations instead of naive $\mathcal{O}(N^2)$.

Remark 2.34. Aliasing: Sampling f at discrete nodes $x_l = \frac{l}{N}$ makes frequencies differing by integer multiples of N indistinguishable, since $\omega_N^{(k+mN)l} = \omega_N^{kl}$. To avoid aliasing high frequencies into lower modes, N must satisfy the Nyquist-Shannon sampling threshold relative to the highest frequency present in f.

Definition 2.35. Trigonometric Interpolating Polynomial & `fftshift`: For N samples $y_l = f(t_l)$ at $t_l = \frac{l}{N}$, the trigonometric polynomial $p_N(t)$ satisfying $p_N(t_l) = f(t_l)$ is expressed in symmetric frequency order:

$$p_N(t) = \sum_{k=-N/2}^{N/2-1} \zeta_k e^{2\pi i k t}$$

Standard FFT outputs store coefficients indexed in non-symmetric order $k \in \{0, \ldots, N-1\}$. Applying `fftshift` rearranges indices from $[0, N-1]$ to symmetric frequency spectrum $[-\frac{N}{2}, \frac{N}{2} - 1]$.

Definition 2.36. Cyclic Convolution: The cyclic convolution $c = a \otimes b$ of two N-periodic vectors $a, b \in \mathbb{C}^N$ is defined as:

$$c_k = \sum_{n=0}^{N-1} a_n b_{(k-n) \pmod N}$$

Theorem 2.37. Discrete Convolution Theorem: The DFT converts cyclic convolution into pointwise vector multiplication:

$$\mathcal{F}_N(a \otimes b) = \mathcal{F}_N(a) \odot \mathcal{F}_N(b)$$

This allows evaluating cyclic convolution in $\mathcal{O}(N \log N)$ operations via:

$$a \otimes b = \text{IFFT}(\text{FFT}(a) \cdot \text{FFT}(b))$$

Linear convolution of length-N sequences can be evaluated using cyclic convolution by zero-padding both sequences to length at least $2N - 1$.

Remark 2.38. Zero-Padding for High-Resolution Evaluation: To evaluate an interpolating trigonometric polynomial $p_N(t)$ at $M \gg N$ fine grid points:

1. Compute $c = \text{FFT}(y)$ of size N and center coefficients using `fftshift`.
2. Pad the frequency vector with $M - N$ zeros at the high-frequency boundaries.
3. Apply `ifftshift` and compute $\tilde{y} = \frac{M}{N}\text{IFFT}(\tilde{c})$ of size M.

This yields exact evaluation of $p_N(t)$ on a fine grid without computing individual exponential sums.